

Unmanned Surface Vehicle for Automatic Water Quality Monitoring

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Abstract. This paper presents an affordable, yet efficient approach to Unmanned Surface Vehicle (USV) design for automated water quality monitoring. Mechanically, the USV employs a catamaran design with robust hardware components, including aluminium structural frames, PVC inflatable tubes, and waterproof enclosures for housing electronics and batteries. The water quality monitoring function is achieved through a multi-parameter sonde controlled by an onboard processing unit, enabling fully automated environmental data collection. With the capability of both GPS-guided and telemetry operation, the USV is suitable to perform various missions/tasks of vast environmental surveillance application on both static water and river setting while maintaining communication with ground control station and server via wireless RF/4G/LTE/GPRS/Wi-Fi networks.

1 Introduction

In the realm of technological innovation, the integration of unmanned systems has revolutionized various fields, and the domain of water quality monitoring is no exception. The advent of Unmanned Surface Vehicles (USVs) has paved the way for a new era in automatic water quality monitoring, combining cutting-edge robotics with environmental science. These advanced vehicles offer unparalleled capabilities for collecting real-time data on various parameters such as temperature, pH levels, dissolved oxygen, and pollutants. Their autonomous navigation systems enable them to traverse water bodies efficiently, covering large areas and reaching remote or hazardous locations that are challenging for manned vessels. Additionally, their ability to operate continuously for extended periods minimizes the need for human intervention, reducing costs, preventing risks, and enhancing data collection reliability. As a result, USVs have become invaluable instruments for researchers, environmental activists, and policymakers striving to understand and protect aquatic ecosystems.

In light of discussion, this paper focuses on the design and implementation of an Unmanned Surface Vehicle (USV) intended for environmental monitoring purposes. Technically, USVs are quite complex in design as they required consistent coordination among mechanical components, sensors, actuators, and navigation algorithms to effectively figure out their designated missions. The mechanical configuration of a USV is identical to a conventional craft, however, smaller in size [1]. There is a variety of vessel designs, in which mono-hull and catamaran (dual hull) design are the most prevalent [10]. The catamaran design provides enhanced stability and payload capacity compared to mono hull designs, which makes it a popular choice for designing and testing platforms in the research field [4]. Despite the mechanical design, it is also required to build the Power Supply, Guidance, Navigation, Propeller and Control system.

In recent years, several USVs are developed by the research community. These systems are designed based on the specific application to be performed. For instance, Springer 1 [7] is developed with the catamaran design for undertaking pollutant tracking, and environmental and hydrographic surveys in rivers, reservoirs, inland waterways and coastal waters, particularly where shallow waters prevail. Springer 1 also serves as a test bed platform for research in intelligent navigation systems and sensor and instrumentation technology. SCOUT [8] stands as a mono-hull USV, engineered with simple, resilience and affordable platform for refining robust algorithms and control software for Autonomous Underwater Vehicle (AUV). VIAM-USV2000 [9] is an unmanned surface vehicle (USV) designed to perform missions on hydrographic surveys and environmental monitoring. VIAM-USV2000 offers a cost-effective and efficient solution for data collection in marine environments. Various robotic companies have also ventured into the development of commercial USVs. For instance, the Wave Glider, a mono-hull design USV, is created by Liquid Robotics Inc. (<https://www.liquid-robotics.com/>). Similarly, Clearpath Robotics has developed the Kingfisher, which features a catamaran design (<https://clearpathrobotics.com/>). The commercial USVs are well equipped with functionalities. However, they are very expensive, difficult to import and not easy to deploy operations in Vietnam. This leads us to the motivation of developing our own USV for particular automatic water quality monitoring applications which involves an

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affordable mechanical fabrication using PVC tubes as well as efficient control systems, electronics hardware and software relying on off-the-shelf, easy-to-find components. Here, the vehicle is equipped with (1) renewable, rechargeable solar energy system to guarantee its operation over prolonged missions, (2) laboratory and industrially calibrated multi-parameter water quality monitoring sonde which is able to perform reliable measurement, yet at much affordable cost in comparison to commercial counterparts, (3) 4G/RF/LTE/GPRS/Wi-Fi wireless communication systems for seamless data transmission, (4) high-speed 4K on-board camera for surveillance and telemetry operation and (5) ArduPilot-based GPS-guided navigation for automated tasks.

The remainder of this paper is organized as follows. Section 2 describes the hardware (both mechanical and electrical aspects) design and implementation. The software architecture is presented in Section 3. Section 4 touches the design and implementation of an affordable yet efficient multi-parameter water quality monitoring sonde. Section 5 discusses the experimental results of USV's operation on a programmed mission on the Saigon river; and finally Section 6 concludes the paper.

2 Hardware Design and Implementation

The hardware design plays a key role in the vessel's reliability and performance. A well-engineered hardware design ensures the USV's stability, buoyancy, and hydrodynamic efficiency, enabling smooth and efficient navigation in various marine conditions. This section will reveal the USV's design process and core hardware components such as sensors, actuators, computing units and body hull.

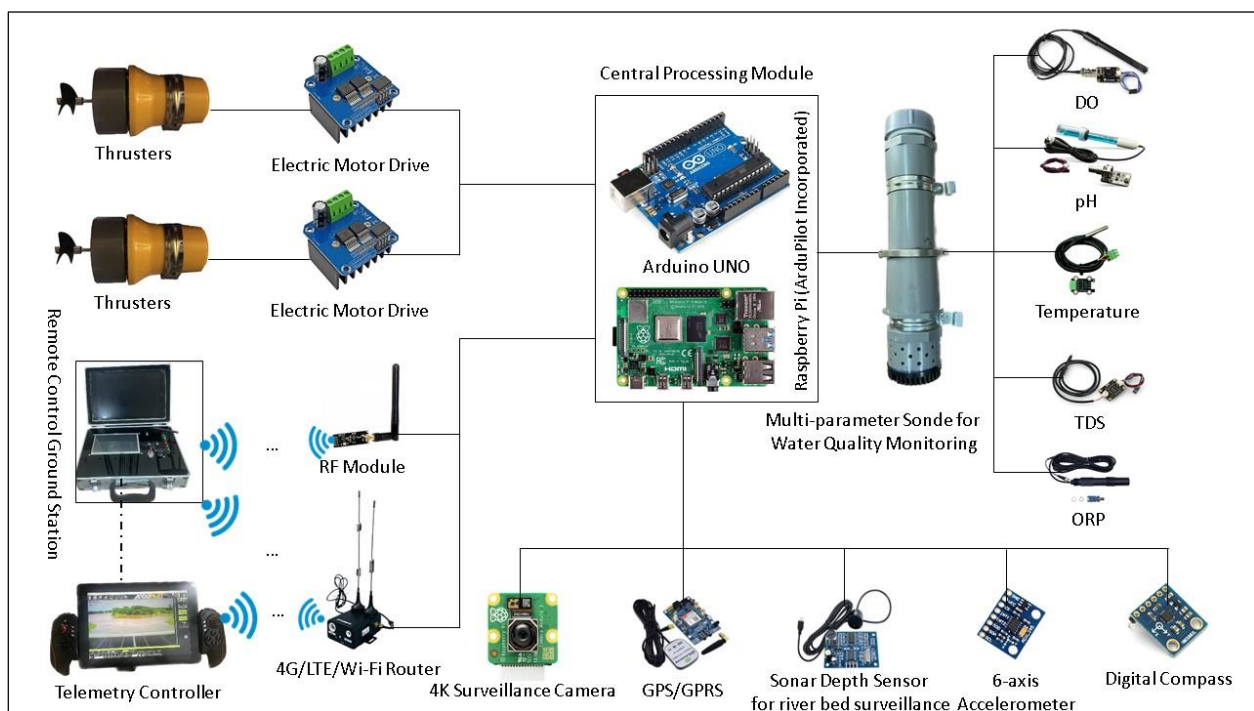


Fig. 1: Schematic of sensors, actuators, and electronic components of the USV

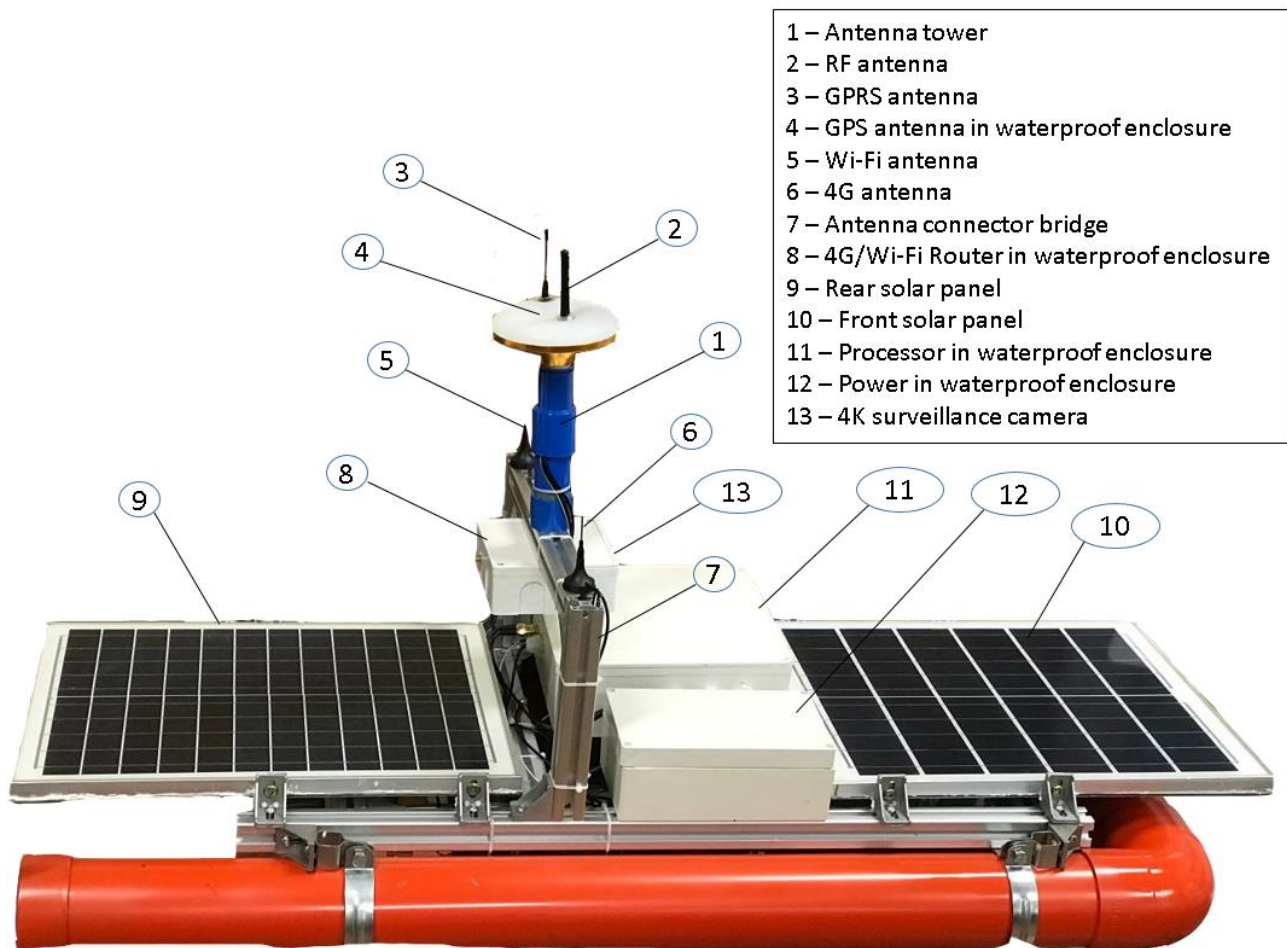


Fig. 2: USV's construction and components

2.1 Electronic components

The main components of the USV are shown in Figures 1-6. The actuation system consists of two OSÁKA DC submersible water pumps which were converted into thrusters by removing the protective cover and modifying the water intake propellers into clockwise and counter clockwise water-pushing propellers. The propeller's maximum rotational speed is around 4800 rpm and it can produce up to 70L/min thrust. The rpm is controlled using a power amplifier integrated circuit, namely BTS7960. This motor driver utilizes duty cycle of PWM signal to control the speed as well as the direction of the thrusters. The control input of BTS7960 is transmitted from an Arduino UNO board designed by Interaction Design Institute Ivrea (<https://www.arduino.cc/>). This board interfaces with a SIM808 module to receive GPS data pertaining to the vehicle's coordinates, and an NRF24L01 module to receive RF signals regarding to the remote control. In autopilot mode via ArduPilot, the Arduino UNO continuously transmits the GPS data to the Raspberry Pi 4 model B to calculate the USV's position and heading to navigate it along a predefined route. In manual mode, the operator will interact with the NRF24L01 module to send control commands to the Arduino UNO, thereby controlling the speed and direction of the vessel. The Arduino UNO acquires water quality data measurements from environmental sensors mounted inside the USV's sonde and depth data measurements from ultrasonic sensor mounted on the USV's hull for water bed mapping. The Raspberry Pi 4 model B (as well incorporated with ArduPilot program) is connected to an Industrial 4G/Wi-Fi Router H927 and a 4K surveillance camera placed within a waterproof enclosure. It is responsible for bidirectional data transmission between the vessel and the Remote Control Ground Station, as well as acting as the ears and eyes of the operator when the vessel operates beyond line-of-sight.

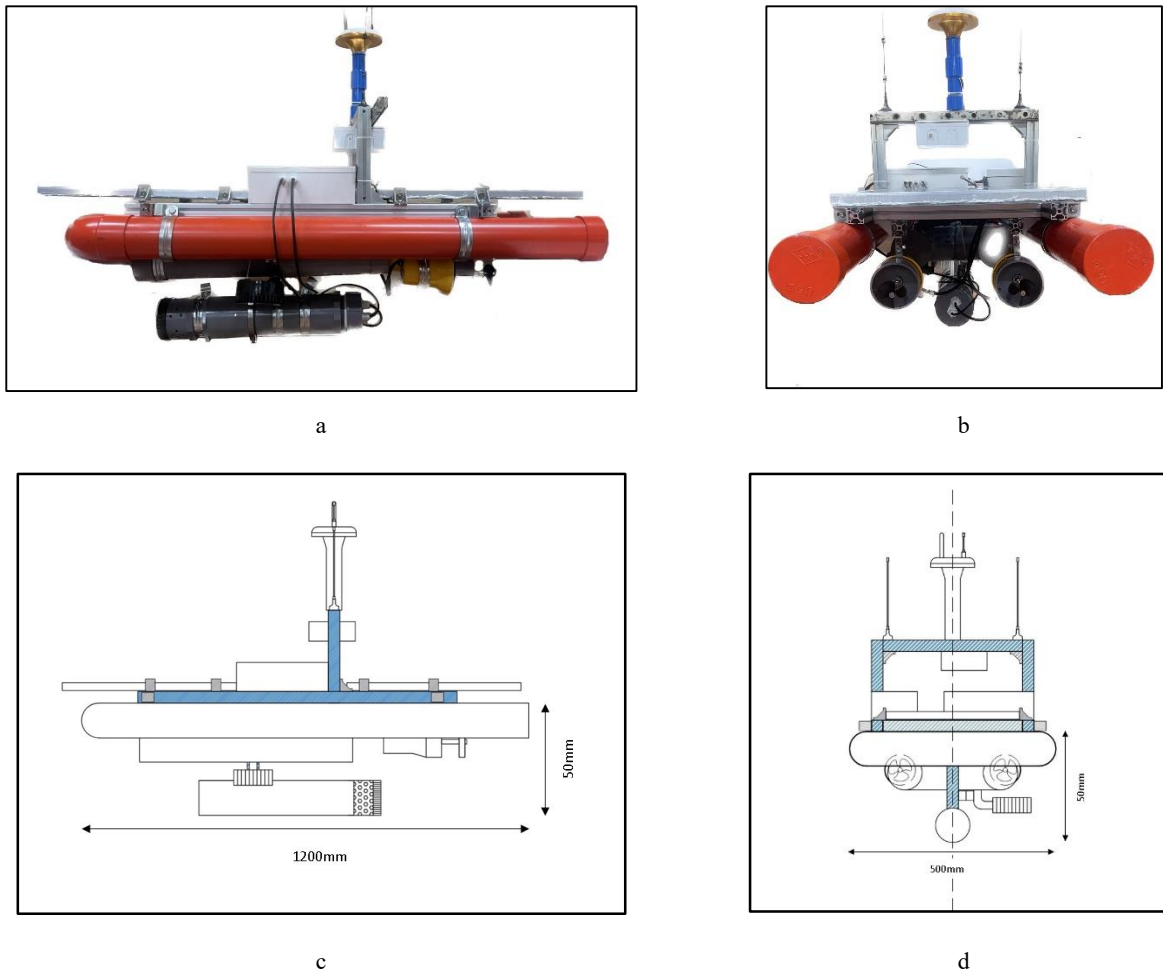


Fig. 3: Hull structure of the USV. a) Side-view image of the USV. b) Rear-view image of the USV. c) Exploded side-view of the USV. d) Exploded rear-view of the USV. The geometric dimension of the vehicle (Width x Length x Depth) is at 500mm x1200mm x50mm with the designed capability to carry a load of up to 21kg



Fig. 4: USV's Remote Control Ground Station with telemetry controller

2.2 Mechanical components

The USV employs the structure of catamaran design as it is evidently better stability and capability to carry heavy payload over the mono-hull design for autonomous vessel. The hull structure is assembled from aluminium structural frames, forming a rectangular shape as shown in Fig. 3. Along both sides and frontal section of the USV are inflatable tubes made from PVC (Polyvinyl Chloride) material, which is widely acceptable for its low cost and availability. These inflatable tubes are assembled onto the aluminium structural frame through stainless steel hanging clamps and are painted in bright orange to make the USV stand out against the water background. The payload of the USV is 21kg and it can move with the speed of up-to 8 knots. Thrusters are also connected with the body hull using stainless steel hanging clamps. Three separate water tightening enclosures are used to carry all the electronic components such as Raspberry Pi 4 model B,

Arduino UNO, sensor circuits, wireless router, and batteries (Figure 5). These enclosures are connected with the frame using nuts and bolts.



Fig. 5: Control electronics compartment

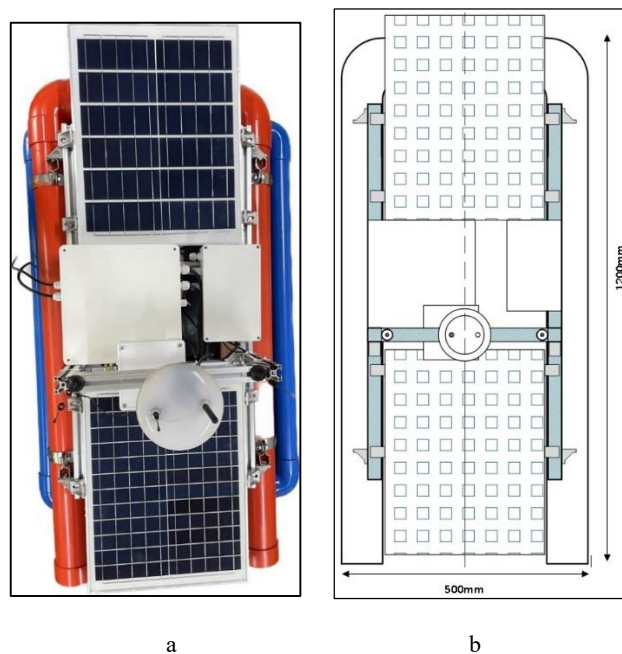


Fig. 6: The installation of solar panels on the USV. a) Top-view image of the USV. b) Exploded top-view image of the USV

2.3 Power supply

To sustain the USV's operations, a high-capacity 12V 10000mAh Lithium battery is employed as shown in Fig. 5. Besides, to ensure the maintenance of energy for long voyages, the USV is equipped with two separate solar panels of moderate size mounted on the front and rear sections of the hull (Fig. 6). The dimensions of each solar panel are 450mm x 350mm with a voltage of 12-24V and a power rating of 40W. These solar panels can provide supplemental energy to the Lithium batteries during the daytime to help endure the operation of the vessel.

3 Software Architecture

The software architecture of the USV originates from the work [11], as depicted in Fig. 7. The main software system resides in the Arduino UNO board, it consists of the USV Firmware, which provides sensor and actuator interface for the high-level application. Other communication and live streaming softwares are running on the Raspberry Pi 4 model B that has more RAM and powerful processor for fast processing.

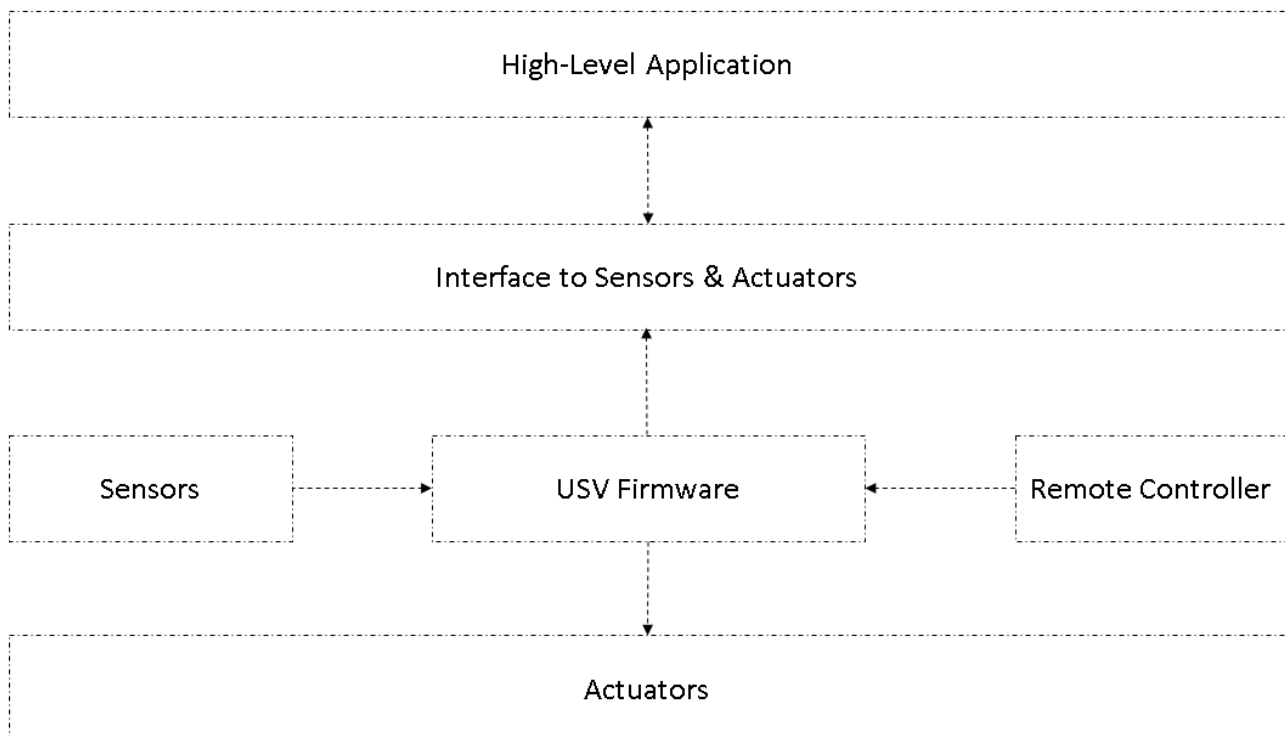


Fig. 7: Overview of software architecture of the USV

3.1 USV Firmware

USV firmwares are C++ based software applications, developed for both Arduino UNO and Raspberry Pi 4 model B on top of Raspberry Pi operating system which is a Linux-kernel based OS. The Arduino Firmware module contains low-level drivers to communicate with GPS/GPRS module, actuators (such as thrusters), and provide message protocol to send and receive data from the sensors and actuators. Moreover, Arduino Firmware is also responsible for receiving the commands from the manual remote control via NRF24L01 and forwarding them to BTS7960 to control the thrusters for telemetry operation (structured as shown in Fig. 8). The Raspberry Pi Firmware includes the following modules:

- Communication module is in charge of transmitting online data from the USV to the central control server via the 4G broadband cellular network. The data includes water quality parameters sent directly from the Arduino Firmware through serial communication protocol, as well as real-time image data from the surveillance camera.
- In autopilot mode of operation, ArduPilot program is incorporated in order to coordinate the GPS guided navigation and path planning of the vehicle over the assigned missions/tasks.

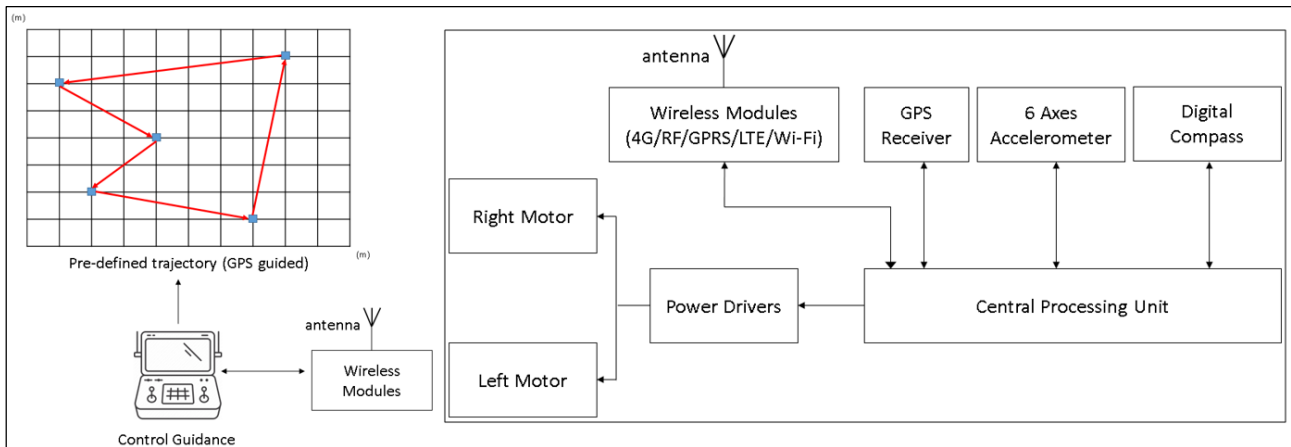


Fig. 8: USV trajectory (GPS guidance) tracking and operational structure

3.2 Remote Control Ground Station

The Remote Control Ground Station (Fig. 4) of the USV is a system comprising web and Windows applications hosted on a server running on top of the Windows operating system. The web applications are developed using the C# programming language and utilizing Microsoft's ASP.NET Core 8.0 technology with the MVC (Model-View-Controller) architectural design pattern. The main applications of the Control Ground Station include:

- A Web-based digital map application which is developed base on Web-GIS technology. This web application provides an intuitive interface comprising a digital map to display geographical data and visualise water quality of the monitored area. In addition, the application also includes a control panel with buttons, sliders and a live view which allows users to directly interface and remote control the USV via live streaming images from its surveillance camera
- A Web API application providing HTTP protocols such as GET, POST, and PUT to obtain data from the USV and store into database
- A Database which is created and managed using the Microsoft SQL Server enabling it to organize and store data related to geography such as layers, colors, makers as well as water monitoring data, and data regarding the coordinates and status of the USV
- Operational commands and relevant hardwares which are bi-directionally communicated with the USV via 4G/RF/LTE/GPRS/Wi-Fi networks to control the USV during its missions in manual, telemetry and autopilot modes of operation.

4 Multi-parameter Water Quality Monitoring Sonde

To implement the environmental monitoring function of the USV, we developed a multi-parameter water quality sonde. This is a completely independent module mounted underneath the hull of the USV and controlled by the USV's on-board processor, as well as being supplied with the power from the USV's batteries. The sonde has a cylindrical structure and is made of waterproof PVC material, as shown in Fig. 9. The head of the sonde has 4 probes for sensing water quality parameters such as pH, temperature, TDS, ORP, and DO as shown in Fig. 9-b. The tail of the sonde has 2 waterproof connectors for wiring the sensors from the sonde to the main waterproof enclosure of the vessel. When powered, the sonde will continuously sample the water quality parameters and send the values to the Arduino. The Arduino integrates all the sensor data and send to the Raspberry Pi to post back to the Remote Control Ground Station.

Here, this sonde has been calibrated accordingly to the environment of operation (PH level of 5.0 to 8.0) using industry graded instrument (HI98194 Hanna Instruments Sonde).

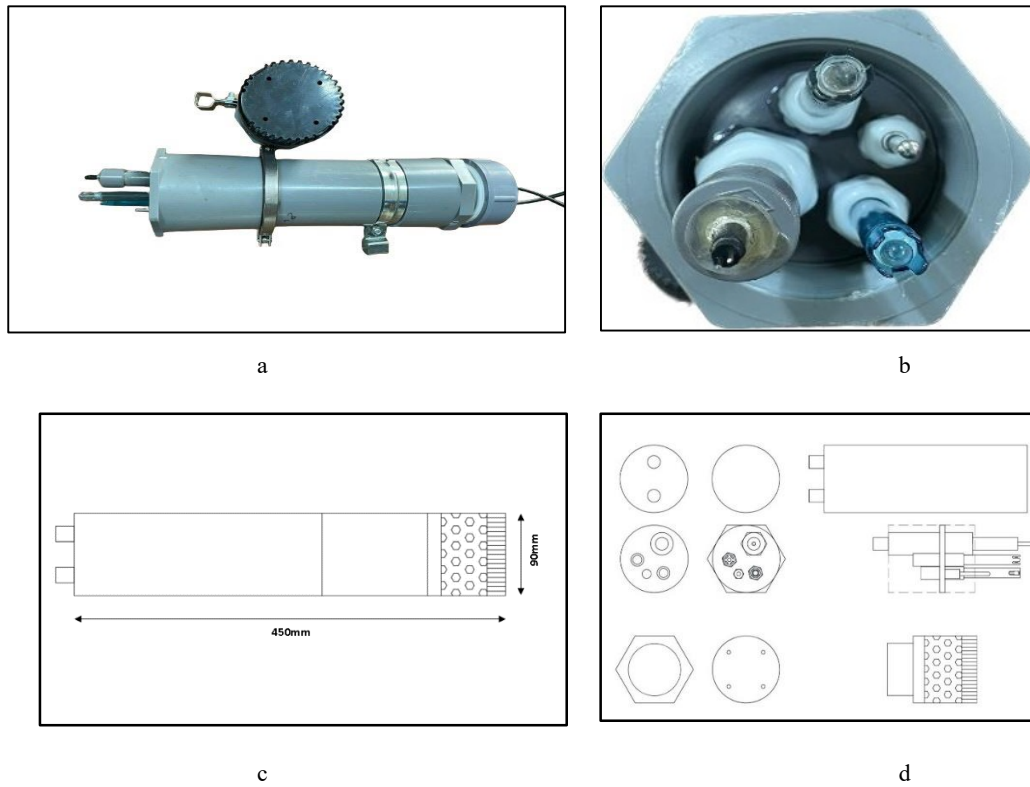


Fig. 9: The USV's sonde for water quality monitoring. a) Full-body image of the sonde and sonar depth sensor. b) Image of the front section of the sonde. c) Cross-sectional drawing of the sonde. d) Cross-sectional drawing of the front and rear section of the sonde



Fig. 10: Snapshot of USV's experiment on Saigon river

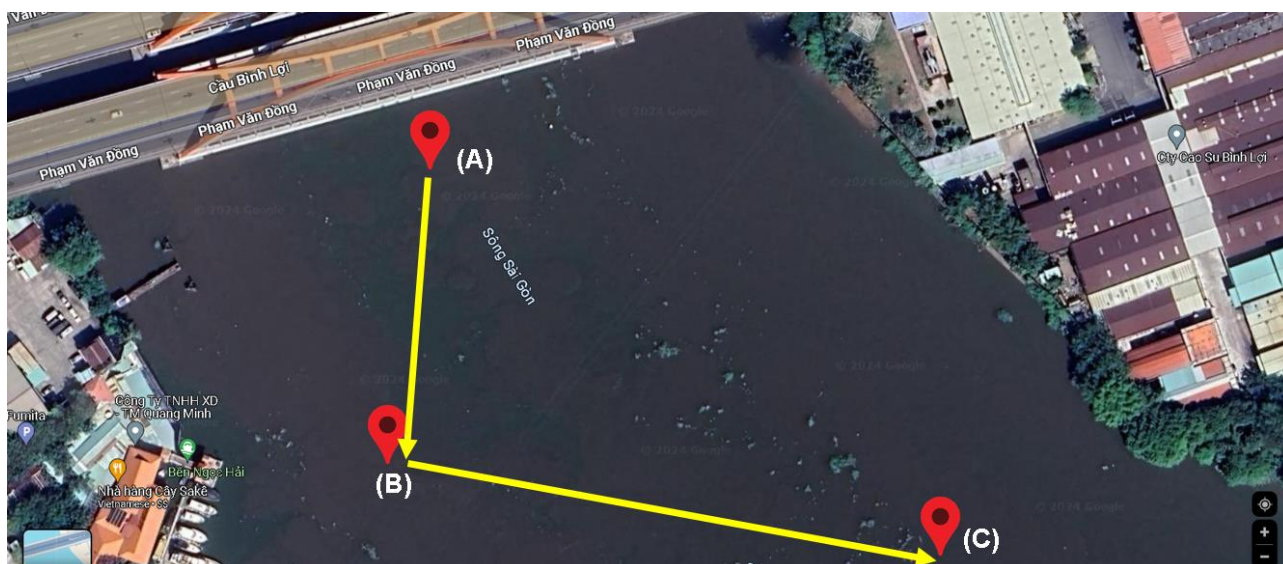


Fig. 11: USV's experiment in autopilot mode through predefined waypoints

5 Experimental Results

We conducted comprehensive tests to validate the USV's design and manoeuvring capability in a swimming pool with no waves and streams. Moreover, to evaluate the operational capability of the USV in realistic conditions, we also conducted extensive tests on a river. Figure 10-11 show snapshot of the USV operating on the Saigon river through 3 waypoints with coordinates A(10.824186, 106.709714), B(10.823301, 106.709724), and C(10.823027, 106.711288). In the manual mode, the USV shows precise control in the swimming pool and it also performed well on the river with the presence of small waves and light winds. When encountering large waves, the USV may experience slight tilting and difficulty in navigation. However, it still retains the stability and balance on the water surface after the waves have passed. In autopilot mode, the USV's trajectories have been programmed for each GPS coordinates and tasks. The real-time video surveillance work seamlessly, and USV operational data as well as sonde measurement results have been continuously updated to the Remote Control Ground Station via 4G/LTE network.

Here, in order to validate the merits of USV measurement, samples were manually collected at programmed locations and post-operationally verified using a Hanna Instruments sonde. The results showed strong correlation with the USV sonde's measurements, with pH, TDS, DO and ORP measurements exhibiting a $\pm 5\%$ tolerance in comparison to the in-laboratory results obtained with the Hanna Instruments HI98194, as illustrated in Table 1.

Table 1: Experiment results of USV's measurement in Saigon river

Waypoint	USV's Multi-Parameter Sonde					In-laboratory Hanna Instruments HI98194				
	T(°C)	pH	TDS	DO	ORP	T(°C)	pH	TDS	DO	ORP
A(10.824186, 106.709714)	28.71	5.81	2453	0.15	300.6	-	5.73	2464	0.12	295
B(10.823301, 106.709724)	28.92	5.62	2500	1.8	309.3	-	5.39	2462	1.77	326.8
C(10.823027, 106.711288)	28.83	5.46	2486	1.79	327.6	-	5.21	2440	1.88	315.9

6 Conclusion

This paper has documented a low cost yet efficient unmanned surface vehicle to the problem of automated water quality monitoring and surveillance. The vehicle is capable to operate in a variety of modes: manual operation via RF, telemetry operation via real-time streaming video and 4G/LTE network as well as automated GPS-guided operation via ArduPilot. The in-house fabricated multi-parameter water quality sonde performs as designed; and the experimental results have proven the merit of such an approach as well as demonstrated its huge potential in marine robotics applications for environmental monitoring and surveillance.

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